Daniel Arbeláez Álvarez

Software Engineer · University Lecturer

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Supporting documents

PROFILE

Software engineer and university lecturer. Employee of the Judicial Branch of Colombia since 2014. In a personal capacity, I develop free software that supports document management for electronic court case files, designed for operational tasks: it does not weigh evidence or take part in judicial decisions. I was a member of team Marduk, winner of the Data-Driven Justice category of JusticIALab 2024. I have published on LegalTech in Cuaderno Activa (2024) and I am studying for the Specialization in Artificial Intelligence.

Technologies: Python (Streamlit, FastAPI, pandas, spaCy) · Language-model APIs (OpenAI, Anthropic) and RAG · Docassemble · Docker · PostgreSQL and SQLite · R · Power BI · Git and GitHub.

EDUCATION

Specialization in Artificial Intelligence · Fundación Universitaria Internacional de La Rioja (UNIR Colombia) — In progress, since 2025

Software Engineering · Tecnológico de Antioquia — 2021

Thesis project: web application for monitoring and managing air-quality information in Segovia, Antioquia. (institutional repository, in Spanish)

EXPERIENCE

Judicial Branch of Colombia · Employee — November 2014 – present

Since 2024, technology support and judicial information systems.

Tecnológico de Antioquia · Adjunct lecturer — February – November 2024

Backup Tecnología (Medellín) · Technical support assistant — March 2012 – July 2014

Preventive and corrective maintenance of computers and printers; software installation and updates; support for voice and data networks.

LEGAL SOFTWARE DEVELOPMENT

AgilEx (formerly GestionExpedienteElectronico) — Since 2022 · MIT · Repository

Creates and fills in the electronic index of a court case file from the metadata of the files in a folder (Agreement PCSJA20-11567 of 2020).

What it does not do: Works only with metadata (name, dates, pages, format and size): it does not assess document content or take part in judicial decisions.

Supporting documents: First commit: 5 January 2022. DNDA copyright registration: 3 November 2022 (Book 13, Volume 92, Entry 259), version 1. 9 published releases and 2,426 downloads on GitHub (October 2026).

TYBABot — 2025 · MIT · Repository

Conversational assistant that guides users on Justicia XXI Web (TYBA). Built from a template by another author.

What it does not do: It does not access Justicia XXI Web or case files; the demo uses third-party AI services.

DisciplinaJudicialAI — 2026 · MIT · Repository

Conversational assistant that guides judicial employees on the disciplinary regime. Built from a template by another author.

What it does not do: It does not handle specific cases or replace advice from a lawyer.

Docassemble Expert — 2025 · MIT · Repository

Conversational assistant for writing and debugging Docassemble YAML interviews. Built from a template by another author.

What it does not do: It does not produce legal documents or replace review by a lawyer.

TutelaBot (adaptation) 2025 · MIT (original author) · Repository

Assistant on the acción de tutela and the right of petition, based on the 1991 Constitution, Decree 2591 of 1991 and Law 1755 of 2015. Adaptation of an MIT project by another author.

What it does not do: It provides guidance; it does not replace advice from a lawyer.

Other public repositories: statistical analysis of experimental data in R (2024); chat interface for the Claude and GPT APIs (2025); data extraction with the Spotify API (2022); data analysis with Power BI (2022); visualization of PM10 particulate-matter data in Python (2020). github.com/HammerDev99.

RECOGNITION

2024. Member of team Marduk, winner of the 1st Innovation Contest of the Judicial Branch – JusticIALab 2024, Data-Driven Justice category. Supporting documents: Superior Council of the Judiciary news, 13 December 2024, in Spanish; UTDI certification letter, 14 May 2025, in Spanish.

2024. Mention of Recognition from the Superior Council of the Judiciary to the Marduk project, of which I was a member, for participation in court modernization and systematization programmes (Agreement PCSJA24-12237 of 6 December 2024).

2023. AgilEx (then «LAW») was included in the 2023 Legal Innovation Map of Colombia (Legal Hackers Colombia and ALTCO), Document and Process Automation category. Supporting documents: map excerpt, in Spanish.

2015. Student representative on the Faculty of Engineering Council, Tecnológico de Antioquia.

PUBLICATIONS

Arbeláez Álvarez, D. (2024). Entrevistas guiadas para la optimización de procesos judiciales: un enfoque desde el LegalTech [Guided interviews for optimizing court procedures: a LegalTech approach]. Cuaderno Activa, 16(1), 37–47. doi:10.53995/20278101.1858

TALKS

2024. Shared talk "National case studies: Colombia's experiences incorporating AI into the justice system", in which I presented the Marduk case. Academic Conference on Artificial Intelligence and Its Impact on Justice, Medellín, 18 October 2024. (programme, in Spanish)

2013–2014. Speaker at the 12th and 13th Regional Meetings of Research Seedbeds (Corporación Universitaria Lasallista, 2013; Politécnico Colombiano Jaime Isaza Cadavid, 2014) and at the Tecnológico de Antioquia Research Days (2013).

DIPLOMA PROGRAMMES

2025 · Management Skills and Leadership · Fundación Universitaria Internacional de La Rioja (UNIR Colombia) · 96 h
2025 · Soft Skills · Universidad Sergio Arboleda · 80 h
2024 · Citizen Data Science · Escuela Superior de Administración Pública (ESAP) · 120 h
2024 · Competency-Based Pedagogy (online) · Politécnico Mayor · 120 h
2023 · Computer Crimes, Cyber Risks and Digital Evidence Handling · Universidad Externado de Colombia · 96 h

FURTHER TRAINING

2026 · Artificial Intelligence in Judicial Decisions: Case-Law Analysis and Criteria for Applying the Law · Escuela Judicial Rodrigo Lara Bonilla · 5 h
2026 · Fundamentals of Generative Artificial Intelligence (postgraduate level) · Escuela de Posgrado Newman · 80 h
2025 · Microsoft .NET + Angular training cycle · Escuela Judicial Rodrigo Lara Bonilla e Intelligent Training · 48 h
2025 · Artificial Intelligence Bootcamp, intermediate level · MinTIC – Talento Tech (Universidad de Antioquia y Universidad de Caldas) · 159 h
2025 · Introduction to Public Innovation · Escuela Superior de Administración Pública (ESAP) · 40 h
2024 · Integrity, Transparency and Anti-Corruption · Función Pública · 20 h
2023 · AI Project Formulation for the Justice Sector · Universidad de los Andes · 30 h
2022 · Information Security Fundamentals · Tecnológico de Antioquia · 36 h

LANGUAGES

Spanish (native); English A2 (technical reading).

Supporting documents at portfolio.sprintjudicial.com/en/soportes — updated October 2026.
Personal opinions; I do not represent the Judicial Branch.